

27. Answer: D

Option A: Incorrect. When fast-moving electrons are directed on a metal surface, x-ray photons are emitted, not that of visible light.

Option B: Incorrect. Infra-red radiation's photon energy is too low to have any effect on the metal.

Option C: Incorrect. Microwave's photon energy is far below that of the work function of metals.

Option D: Correct. Photoelectric effect refers to the emission of electrons from a metal surface when electromagnetic radiation of a frequency above the threshold frequency is incident on it.